Math for CS & AI: Homework 1

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Problem 1

Prove that if $f(n), g(n) \geq 0$ for all n, then $\mathcal{O}(f(n)) + \mathcal{O}(g(n)) = \mathcal{O}(f(n) + g(n))$.

Answer:

For an arbitrary function $\alpha \in \mathcal{O}(f)$, there exist constants $C_\alpha > 0$ and $N_\alpha \in \mathbb{N}$ such that $|\alpha(n)| \leq C_\alpha f(n),\ \forall n \geq N_\alpha$. Similarly, for $\beta \in \mathcal{O}(g)$, there exist constants $C_\beta > 0$ and $N_\beta \in \mathbb{N}$ such that $|\beta(n)| \leq C_\beta f(n),\ \forall n \geq N_\beta$.

Let $C = \max(C_\alpha, C_\beta)$ and $N = \max(N_\alpha, N_\beta)$. For all $n \geq N$, we have:

$$\begin{aligned} |\alpha(n) + \beta(n)| &\leq |\alpha(n)| + |\beta(n)| \\ &\leq C_\alpha f(n) + C_\beta g(n) \\ &\leq C\left(f(n) + g(n)\right). \end{aligned}$$

This shows that $\alpha(n) + \beta(n) \in \mathcal{O}(f(n) + g(n))$. Therefore, $\mathcal{O}(f(n)) + \mathcal{O}(g(n)) = \mathcal{O}(f(n) + g(n))$.

Problem 2

Prove that if $f(n), g(n) \geq 0$ for all n, then $\mathcal{O}(f(n)g(n)) = f(n) \cdot \mathcal{O}(g(n))$.

Answer:

For an arbitrary function $\alpha \in \mathcal{O}(f \cdot g)$, there exist constants $C_\alpha > 0$ and $N_\alpha \in \mathbb{N}$ such that $|\alpha(n)| \leq C_\alpha f(n)g(n) = f(n) \cdot C_\alpha g(n),\ \forall n \geq N_\alpha$.

This implies $\alpha(n) \in f(n) \cdot \mathcal{O}(g(n))$. Therefore, $\mathcal{O}(f(n)g(n)) = f(n) \cdot \mathcal{O}(g(n))$.

Problem 3

Prove that if $g(n) > 0$ for all $n \in \mathbb{N}$, then $f(n) = \mathcal{O}(g(n))$ for all $n \in \mathbb{N}$ is equivalent to $f(n) = \mathcal{O}(g(n))$ as $n \to \infty$.

Answer:

On one hand, if $f(n) = \mathcal{O}(g(n))$ for all $n \in \mathbb{N}$, then there exists a constant $C > 0$ such that $|f(n)| \leq Cg(n), \forall n \in \mathbb{N}$. This implies $|f(n)| \leq Cg(n), \forall n \geq 0$, which leads to $f(n) = \mathcal{O}(g(n))$ as $n \to \infty$.

On the other hand, if $f(n) = \mathcal{O}(g(n))$ as $n \to \infty$, then there exists constants $C > 0$ and $N \in \mathbb{N}$ such that $|f(n)| \leq Cg(n), \forall n \geq N$. Since N is a constant, we can define

$$C' = \max\left(C, \max_{n=0}^{N-1} \frac{|f(n)|}{g(n)}\right),$$

which is also a positive constant provided $g(n) > 0, \forall n \in \mathbb{N}$. Then, $|f(n)| \leq C'g(n)$ holds for all $n \in \mathbb{N}$, which means $f(n) = \mathcal{O}(g(n))$ for all $n \in \mathbb{N}$.

Problem 4

Let $f : D \to \mathbb{R}$ be an arbitrary function defined on a domain D with finite number of elements. Show that $f(n) = \mathcal{O}(g(n))$ for all $n \in D$, as long as the function $g : D \to \mathbb{R}$ satisfies $g(n) > 0$ for all $n \in D$.

Answer:

Since D is a finite set, we can list its elements as $n_1, n_2, \ldots, n_k$, where k is the size of D. Define the constant C as follows:

$$C = \max_{i=1}^{k} \frac{|f(n)|}{g(n)}.$$

Given that $g(n) > 0, \forall n \in D$, the constant C is well-defined and positive.

Therefore, $\forall n \in D, |f(n)| \leq Cg(n)$, which implies $f(n) = \mathcal{O}(g(n))$ for all $n \in D$.

Problem 5

Prove that the two definitions (Definitions 2.6 and 2.24) of big-$\mathcal{O}$ notation are equivalent when $g : \mathbb{N} \to \mathbb{R}$ is a function that is positive for all $n \in \mathbb{N}$.

Answer:

First, the following inequality always holds where $g(m) > 0$ for all $m \geq n$:

$$\sup_{m \geq n} \frac{|f(m)|}{g(m)} \geq \sup_{m \geq n+1} \frac{|f(m)|}{g(m)}. \tag{1}$$

On one hand, given constants $C > 0$ and $N \in \mathbb{N}$ such that $|f(n)| \leq Cg(n), \forall n \geq N$, we can conclude that

$$0 \leq \sup_{m \geq n} \frac{|f(m)|}{g(m)} \leq C, \quad \forall n \geq N. \tag{2}$$

Combining (1) and (2), we see that the sequence $\left\{\sup_{m \geq n} \frac{|f(m)|}{g(m)}\right\}_{n \in \mathbb{N}}$ is monotonically non-increasing and bounded below by 0. This implies that

$$\limsup_{n \to \infty} \frac{|f(n)|}{g(n)}$$

exists and is finite. Therefore, the first definition of big-$\mathcal{O}$ notation implies the second one.

Problem 5 continued on next page...

On the other hand, suppose that $g(n) > 0$ for all sufficiently large n and that

$$\limsup_{n\to\infty} \frac{|f(n)|}{g(n)} < +\infty.$$

This means that the limit superior of the sequence equals a certain constant $C' \geq 0$. By the definition of the limit, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left| \sup_{m\geq N} \frac{|f(m)|}{g(m)} - C' \right| < \varepsilon,$$

which implies that for all $n \geq N$,

$$|f(n)| \leq (C' + \varepsilon)g(n).$$

Thus, the second definition of big-$\mathcal{O}$ notation implies the first one.

In conclusion, we have shown that the two definitions of big-$\mathcal{O}$ are equivalent.

Problem 6

Prove that $\sum_{k=1}^{n} k^2 = \Theta(n^3)$.

Answer:

On one hand, for all $n \geq 1$,

$$\begin{aligned}\left|\sum_{k=1}^{n} k^2\right| &\leq \sum_{k=1}^{n} n^2 \\ &= n \cdot n^2 \\ &= n^3,\end{aligned}$$

which implies $\sum_{k=1}^{n} k^2 = \mathcal{O}(n^3)$.

On the other hand, for all $n \geq 1$,

$$\begin{aligned}\left|\sum_{k=1}^{n} k^2\right| &\geq \sum_{k=\lceil \frac{n}{2} \rceil}^{n} k^2 \\ &\geq \frac{n}{2} \cdot \left(\frac{n}{2}\right)^2 \\ &= \frac{1}{8}n^3,\end{aligned}$$

which implies $\sum_{k=1}^{n} k^2 = \Omega(n^3)$.

Combining both bounds, we conclude that $\sum_{k=1}^{n} k^2 = \Theta(n^3)$.

Problem 7

Prove that for n aliens living on a planet with m days in a year, for all $n \geq 1$ as $m \to \infty$, the collision probability is

$$\Pr[A] = 1 - e^{-N/m} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right),$$

where $N := n(n-1)/2$.

Answer:

Let $c = \log_n m$ so that $n = m^c (c \geq 0)$.

For the case that $0 \leq c < \frac{2}{3}$,

$$\begin{aligned}
\ln \Pr[\overline{A}] &= \sum_{k=0}^{n-1} \ln\left(1 - \frac{k}{m}\right) \\
&= \sum_{k=0}^{n-1} -\frac{k}{m} + \mathcal{O}\left(\frac{k^2}{m^2}\right) && \left(\text{since } \frac{k}{m} \to 0\right) \\
&= -\frac{1}{m}\sum_{k=0}^{n-1} k + \mathcal{O}\left(\frac{1}{m^2}\sum_{k=0}^{n-1} k^2\right) \\
&= -\frac{N}{m} + \mathcal{O}\left(m^{3c-2}\right), \\
\Pr[\overline{A}] &= \exp\left(-\frac{N}{m}\right) \cdot \exp\left(\mathcal{O}\left(m^{3c-2}\right)\right) \\
&= e^{-N/m}\left(1 + \mathcal{O}(m^{3c-2})\right) && \left(\text{since } m^{3c-2} \to 0\right) \\
&= e^{-N/m} + \mathcal{O}\left(e^{-n^2/m} \cdot \frac{n^3}{m^2}\right).
\end{aligned}$$

Analyzing the part inside the big-$\mathcal{O}$ notation, we treat n as the variable and compute its derivative. The result is

$$e^{-n^2/m} \cdot \frac{n^2}{m^2}\left(3 - \frac{2n^2}{m}\right),$$

which implies it reaches its maximum when $n = \Theta(\sqrt{m})$. Thus, $\Pr[\overline{A}] = e^{-N/m} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$. Substituting the result for $\Pr[\overline{A}]$ into the expression for $\Pr[A]$ completes the proof.

Problem 7 continued on next page...

For the case that $\frac{2}{3} \leq c \leq 1$,

$$\begin{aligned}
\ln \Pr[\overline{A}] &= \sum_{k=0}^{n-1} \ln\left(1 - \frac{k}{m}\right) \\
&\leq \sum_{k=0}^{n-1} -\frac{k}{m} \qquad \text{(since } \ln(1+x) \leq x\text{)} \\
&= -\frac{N}{m}, \\
\Pr[\overline{A}] &\leq e^{-N/m}, \\
\Pr[A] &\geq 1 - e^{N/m}.
\end{aligned}$$

Since $1 - e^{-N/m} \leq \Pr[A] < 1$, the error of the estimate is $\mathcal{O}\left(e^{-m^{2c-1}}\right)$, which declines faster than the power function $m^{-1/2}$. Therefore, the proof is completed.

For the remaining case where $c > 1$ (i.e., $n > m$), $\Pr[A]$ is 1 by the pigeonhole principle, and the theorem remains valid for the same reasoning.

In conclusion, all the cases of the theorem have been proven.

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Problem 8

For the alien birthday paradox, derive the probability of the following event: there are exactly $n-2$ distinct birthdays among n aliens, where exactly one birthday is shared by three aliens. Then, approximate this probability as a simple function of n and m with error $o(1)$ for all $n \geq 1$ as $m \to \infty$.

Answer:

Let A be the event described in the problem, and $c = \log_n m$ such that $n = m^c (c \geq 0)$.

For every scenario satisfying the constraint, we can view the three aliens sharing a birthday as forming a single group. Therefore, the event can be divined into two parts, that is selecting 3 of the n aliens to have the same birthday and requiring the remaining $n-3$ aliens have unique birthdays in the rest $m-1$ days.

The probability of the three aliens sharing one birthday (event B) is given by

$$\Pr[B] = \frac{1}{m^2},$$

which represents two of them share birthday with the rest one.

The probability of the remaining $n-3$ aliens have unique birthdays in the rest $m-1$ days (event C) has been solved in the last problem, which is

$$\Pr[C] = e^{-(n-3)(n-4)/2(m-1)} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right).$$

Problem 8 continued on next page...

Since the two parts are independent, we can conclude that

$$\begin{aligned}\Pr[A] &= \frac{n(n-1)(n-2)}{6} \cdot \Pr[B] \cdot \Pr[C] \\ &= \frac{n(n-1)(n-2)}{6m^2} \cdot \left(e^{-(n-3)(n-4)/2(m-1)} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right)\right) \\ &= \frac{n(n-1)(n-2)}{6m^2} \cdot e^{-(n-3)(n-4)/2(m-1)} + \mathcal{O}\left(m^{3c-5/2}\right)\end{aligned}$$

For the case that $0 \le c < \frac{5}{6}$, the error $\mathcal{O}(m^{3c-5/2}) = o(1)$, which satisfies the constraint of the problem.

For the case that $c \ge \frac{5}{6}$, we have discussed in the last problem and derived that

$$\Pr[C] = \mathcal{O}\left(e^{-m^{2c-1}}\right).$$

This implies that

$$\Pr[A] = \mathcal{O}\left(\frac{m^{3c-2}}{e^{m^{2c-1}}}\right),$$

which means $\Pr[A] = o(1)$.

In conclusion, the probability of event A is approximately:

$$\Pr[A] = \frac{n(n-1)(n-2)}{6m^2} \cdot e^{-(n-3)(n-4)/2(m-1)} + o(1).$$

Problem 9

Give an example of a function $f : \mathbb{N}^2 \to \mathbb{R}$ such that $f(m,n) = \mathcal{O}(g(m,n))$ as $m, n \to \infty$ for some function $g : \mathbb{N}^2 \to \mathbb{R}$ with $g(m,n) > 0$ but the sum $\sum_{k=1}^{n} f(m,k)$ is not $\mathcal{O}(\sum_{k=1}^{n} g(m,k))$.

Answer:

Consider the following functions:

$$\begin{aligned} f(m,n) &= \begin{cases} 2^m + m + 1, & n = 1, \\ mn + 1, & n \ge 2, \end{cases} \\ g(m,n) &= mn + 1. \end{aligned}$$

It's clear that $|f(m,n)| \le g(m,n)$ holds for all $m, n \ge 2$, which implies $f(m,n) = \mathcal{O}(g(m,n))$. Now, compute the two summations:

$$\begin{aligned} \sum_{k=1}^{n} f(m,k) &= 2^m + n + \frac{mn(n+1)}{2}, \\ \sum_{k=1}^{n} g(m,k) &= n + \frac{mn(n+1)}{2}. \end{aligned}$$

Therefore, $\sum_{k=1}^{n} f(m,k)$ is not $\mathcal{O}(\sum_{k=1}^{n} g(m,k))$.